**Université de Lille**

**IUT de Lille**

**Département informatique**

# Development of an AI for the Penguin Party Game

## Internship Report

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par

Lorenzo FLAGOTHIER

*Encadrant entreprise :* Koji Tajima

*Encadrant universitaire :* Isabelle Delille

**National Institute of Technology Gifu**

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# Table des matières

**Résumé** 1

**Summary** 2

**Introduction** 3

**Internship Context and Presentation** 6

1 Context and Location of the Internship . . . . . 7
2 Working Environment . . . . . 7
3 Technical Framework and Tools Used . . . . . 9
4 Organization of the Project and Internship Goals . . . . . 10

**Technical Work and Implementation** 12

5 Understanding the Game and Defining the Problem . . . . . 13
6 Choice and Implementation of the First AI . . . . . 14
7 Evaluation, Testing, and Refinement . . . . . 15
8 Limits, Learnings, and Future Perspectives . . . . . 16

**Personal and Professional Experience During the Internship** 17

9 Cultural and Social Immersion . . . . . 18
10 Personal Development and Transversal Skills . . . . . 19
11 Professional Impact and Perspectives . . . . . 19

**Conclusion** 21

# Résumé

Ce stage s'est déroulé au National Institute of Technology Gifu, où j'ai travaillé sur le développement d'une intelligence artificielle pour le jeu Penguin Party. Initialement, j'ai conçu et finalisé une IA basée sur l'apprentissage automatique (machine learning). Par la suite, j'ai exploré une approche par Q-learning afin d'améliorer les performances de l'IA, mais j'ai finalement abandonné cette piste. Mon IA actuelle bat des joueurs utilisant une stratégie aléatoire, bien que je n'aie pas encore testé ses performances contre d'autres intelligences artificielles.

Le serveur du jeu a été développé en Java par mon professeur, tandis que les intelligences artificielles ont été programmées en Python, avec l'utilisation de bibliothèques telles que NumPy, pandas, scikit-learn, et joblib. Ce travail m'a permis d'acquérir de solides connaissances en intelligence artificielle et dans son application pratique, notamment dans la conception d'agents capables d'apprendre et de prendre des décisions stratégiques.

Ce projet a été réalisé principalement en autonomie, avec toutefois de nombreux échanges enrichissants avec d'autres étudiants travaillant sur des IA similaires. Enfin, je souhaite, si le temps me le permet, poursuivre le développement de cette IA afin d'en améliorer les performances.

# Summary

This internship took place at the National Institute of Technology Gifu, where I worked on developing an artificial intelligence for the Penguin Party game. Initially, I designed and finalized a machine learning-based AI. Then, I explored a Q-learning approach to improve the AI's performance, but eventually abandoned this path. My current AI beats players using random strategies, although I have not yet tested its performance against other artificial intelligences.

The game server was developed in Java by my professor, while the artificial intelligences were programmed in Python using libraries such as NumPy, pandas, scikit-learn, and joblib. This work allowed me to gain solid knowledge in artificial intelligence and its practical application, especially in designing agents capable of learning and making strategic decisions.

The project was mainly carried out independently, with many valuable exchanges with other students working on similar AIs. Finally, I hope to continue developing this AI to improve its performance if time permits.

# Introduction

As part of the second year of my **Bachelor of Science in Computer Science (BUT)** at the University Institute of Technology (IUT), I had the opportunity to complete a mandatory twelve-week internship from **April 7 to June 30, 2025**. This internship, a key component of my academic program, aimed to immerse me in a real-world professional environment and apply the knowledge and skills acquired throughout my studies. I am proud to have carried out this internship abroad, at the **National Institute of Technology, Gifu College**, a well-respected higher education institution located in Gifu Prefecture, Japan.

This choice was not made randomly. It was a personal goal, deeply connected to my long-standing cultural and linguistic interest in Japan. Having already visited the country once, I was determined to return, this time in a professional context. This internship thus represented a rare and valuable opportunity to combine three major dimensions : **technological learning, personal growth, and international exposure**. It allowed me to test my abilities in a completely unfamiliar environment, while also developing a clearer vision of the kind of career path I want to pursue.

## Internship context and objectives

The subject of my internship focused on a timely and ambitious theme : the **development of an artificial intelligence (AI) for a board game**. The chosen game, *Penguin Party*, is a card-based strategy game in which players must build a pyramid of penguins while anticipating the moves of their opponents. My main objective was to **design and implement a high-performing AI** capable of making consistent and optimized decisions in a constrained, partially observable environment. This project was part of an internal competition organized by the institution, where multiple students competed by developing their own AI agents.

This project presented several layers of complexity. Technically, I had to adapt to a pre-existing

architecture built around a **Java-based client-server system**, with a strict communication protocol governing each move. The AI had to interact with this system while complying with the rules. Moreover, the AI received no information about the other player's cards or decisions, making the development of an effective strategy especially challenging.

This experience also introduced me to advanced AI concepts such as *supervised machine learning*, reinforcement learning (Q-learning), and performance evaluation in probabilistic environments. These areas, which I will explore in detail later in this report, helped me grow significantly in terms of algorithmic creativity, critical thinking, and technical rigor.

## A unique human and cultural experience

Beyond technical skills, this internship was defined by its **rich human and cultural dimension**. From the very beginning, I was warmly welcomed by both the faculty and the Japanese students. My internship supervisor was supportive and approachable, providing just the right balance between guidance and autonomy. I also had the chance to interact with other students working on similar projects, which led to informal test sessions and insightful exchanges.

Throughout the internship, I took part in numerous events organized by the school, such as a large **Sports Day**, a **community-wide barbecue** for all dormitory residents, and a weekly event called *Talk Café*, where international and Japanese students met to play board games and exchange ideas. These events made my integration much easier and allowed me to form strong bonds of friendship with many of the local students. Very quickly, I felt at home, and I developed a comfortable daily routine that combined work, social interaction, and discovery.

Daily life in Gifu, shared with other international students, also played a major role in making this experience unforgettable. My typical day consisted of working from morning until noon, then biking to the local convenience store for lunch, before returning to work in the afternoon. After 4 :10 PM, I often stayed at the campus to chat and play games with the students. In the evening, I occasionally shared meals with them before returning to the dormitory to talk with my friends back in France. These moments of connection, both in **English and Japanese**, helped me improve my language skills and communication abilities in a multicultural context.

## A turning point in my academic and personal journey

This internship marked a decisive step in my personal and professional journey. Not only did it allow me to deepen my understanding of artificial intelligence, but it also opened my eyes to a field that I was not initially interested in. Over time, I began to develop a genuine passion for AI, and I am now considering exploring it further in the future.

More importantly, this internship taught me that I am capable of managing on my own, even in a completely foreign environment. I learned to adapt, to work independently, and to create meaningful relationships across language and cultural barriers. This experience also opened real career possibilities, as I received several job offers in Japan — not necessarily in IT, but nonetheless very motivating and encouraging.

## Structure and purpose of this report

The purpose of this report is to provide a clear and structured overview of my internship experience, highlighting the technical, professional, and personal skills I developed. It is divided into three main parts :

- The first part will introduce the institution, the working environment, available tools, and the overall organization of the project.
- The second part will focus on my main mission : developing an AI for *Penguin Party*. This includes the challenges I faced, the different approaches I tested, and the results obtained.
- The final part will reflect on the broader experience, the skills I acquired, and the perspectives opened by this internship — both professionally and personally.

Through this analysis, I hope to convey the full richness of this journey and demonstrate how it has shaped both my academic growth and personal development.

# Internship Context and Presentation

# 1 Context and Location of the Internship

My internship took place at the **National Institute of Technology, Gifu College** (commonly referred to as *Gifu Kosen*), an engineering school located in **Motosu**, in **Gifu Prefecture**, Japan. I was part of the **Computer Science Department**, and more specifically, I worked in the **laboratory of Professor Koji Tajima**, who was my main supervisor during the internship.

I had the opportunity to go there thanks to an **exchange partnership between the Japanese Kosen system and several universities in the north of France**. While this internship could have been completed in France, I saw it as a unique chance to **develop my skills in an international context**, and to **return to Japan**, a country I already knew and deeply appreciated.

Since it was my second time in Japan, I didn't feel nervous or hesitant before arriving. On the contrary, I felt **rather confident** and eager to reconnect with the Japanese environment and culture.

The welcome at Gifu Kosen was **exceptionally warm and friendly**. Both the teachers and students showed genuine interest in me, which made it very easy to integrate into the daily life of the institution. I truly felt valued and supported, and that encouraged me to give my best throughout the internship.

**Professor Tajima**, beyond being a rigorous academic, is also **a board game enthusiast**, which played a significant role in the choice of the project I was assigned. His lab is not specialized in artificial intelligence, but he still offered me the chance to **work freely on an AI project**, giving me the flexibility and tools I needed to explore the topic in depth. His approach focuses on **autonomy, experimentation, and human-centered guidance**.

This environment, both **socially rich and intellectually stimulating**, allowed me to make great progress while also discovering **a new way of working and thinking**, through the lens of Japanese academic culture.

# 2 Working Environment

The internship took place within the **Department of Computer Science** at the **National Institute of Technology, Gifu College (Gifu Kosen)**, located in Motosu, Japan. This technical college offers a five-year program, with students specializing in a specific field starting from the third year. My internship was made possible thanks to a partnership between several **Japanese Kosen** and universities in the **north of France**.

Although it was not a research laboratory in the strict sense, the workspace I was assigned to was part of the department's infrastructure and supervised by **Professor Koji Tajima**. The environment was **functional, calm, and open**, designed to support students working independently or collaboratively on their projects.

Roughly **ten people** were regularly present in the room : **six Japanese students**, **two other international interns**, and myself, under the supervision of **Professor Tajima**. Each student had their own **dedicated workstation** equipped with a personal computer. The machines were powerful enough to handle modern development tools, allowing me to use everything required for my project : **Java** for server integration and **Python** for AI development and testing.

Internet access and required tools were **freely available**. I had complete freedom to install the necessary libraries, configure my development environment, and test my code without any restrictions. This technical freedom was **crucial to my productivity** throughout the internship.

Despite the language barrier, interactions with the other students were **warm and welcoming**. We were able to exchange ideas, help each other out, and even debug code together when needed. The working atmosphere was both **relaxed and focused**, ideal for autonomous work.

**Professor Tajima** was always **available** to answer questions and offer guidance when needed. He also encouraged communication between students, particularly to help his students practice English. His passion for board games was reflected in the room, which included an **impressive collection of board games**, adding a playful dimension to the workspace.

A key aspect of this internship was the level of **autonomy** granted to me. There was no imposed methodology or fixed roadmap—only a general mission. I had to define my own **workflow**, priorities, and project timeline. This freedom made the experience feel very close to a **real-world professional context**, where personal initiative is essential.

In summary, the environment provided by Gifu Kosen was both **flexible and stimulating**. I had access to everything I needed to work efficiently, in a calm and well-equipped setting, with enough support to avoid feeling isolated, yet enough independence to grow at my own pace.

# 3 Technical Framework and Tools Used

The technical environment during the internship was primarily based on a **Java-based server** developed by Professor Koji Tajima. This server acted as the official platform to simulate *Penguin Party* matches between different artificial intelligences. It strictly enforced the game's rules and followed a precise communication protocol. Each AI connected to the server as a client and could only interact through a defined exchange of messages. This interaction was limited to the current state of the pyramid and the AI's own hand, without any information about the opponent's cards or strategy. This **constraint** made developing a competitive AI particularly challenging, as it had to operate under significant **uncertainty**.

For developing my AI agents, I primarily used **Python**, a language well suited for rapid prototyping and AI development. Although the server was implemented in Java, I created a Python client that respected the server's communication protocol. This approach allowed me to benefit from Python's **flexibility** and **power** while complying with the technical constraints imposed by the Java server.

Several Python libraries played a crucial role throughout the project. **NumPy** was used for numerical computations and array manipulations, particularly to manage game states and probability distributions. **Pandas** helped structure and analyze the data collected during training phases, enabling me to evaluate and compare agent behaviors effectively. The `RandomForestRegressor` from the **scikit-learn** library was central to building supervised learning models for state evaluation, providing robust regression capabilities with excellent generalization. The **joblib** library allowed efficient saving and loading of trained models, facilitating iterative testing without retraining from scratch each time. Additionally, **Matplotlib** was employed to visualize results such as win rates and model performances. The standard Python `random` module was also used for action selection in some strategies.

My daily work routine aligned with the institute's official schedule, from **9 :00 AM to 4 :10 PM**, Monday through Friday. Most of the time was dedicated to laboratory work, where I could code, test, and analyze my AI agents in a quiet and focused environment. While I worked autonomously on my project, I shared the workspace with several Japanese students developing their own strategies for the same competition. This created a **stimulating yet low-pressure atmosphere**. Informal discussions occasionally arose around game tactics or implementation tips, which helped broaden my understanding and improve my approaches. Despite the language barrier, these exchanges happened in English, simple Japanese, or simply through a shared enthusiasm for the project.

Overall, the technical environment was well balanced. I enjoyed the freedom to explore

different ideas, the clear framework of a stable server, and access to tools that enabled rapid iteration. This **autonomy**, combined with occasional interactions with other students, provided ideal conditions to learn and be creative. The laboratory atmosphere was calm and productive, and I always felt **welcomed** and **encouraged** in my work.

# 4 Organization of the Project and Internship Goals

Unlike other internships that may follow a **strict schedule** with intermediate deliverables and a predefined methodology, my internship was conducted in a deliberately **open and flexible framework**. There was no formal planning or imposed work process. The only real constraint was that the project had to be completed by the **end of the internship**. Beyond that, I was completely **free** in how I approached the work — from choosing the type of artificial intelligence to defining the learning process and designing the logic of my algorithms.

The institution provided a shared framework : a **Java-based game server** simulating the board game *Penguin Party*, and the requirement to use either `Java` or `Python` as the programming language. Within this framework, however, I had **complete autonomy** to define my objectives, choose my tools, and organize my workflow. This freedom was both a **challenge** and an **excellent opportunity** to explore AI development on my own for the first time.

I began by **researching various types of artificial intelligence**. As a beginner in the field, I wanted to start with something *accessible yet meaningful*. I initially chose a **machine learning** approach, with a simple goal : to train an AI capable of easily defeating both a human player and a random AI. Once I achieved a satisfactory level of performance, I explored other techniques to compare their effectiveness.

After building my first machine learning model, I tested a **Q-learning** approach, which is a form of *reinforcement learning*. However, this attempt was not as successful — the resulting agent performed worse than my initial model. I eventually returned to my first approach and continued refining it. This **trial-and-error** process confirmed the importance of **experimentation** in AI development.

My daily routine was **self-directed**. I worked on-site from morning to late afternoon, developing on my own machine and interacting with the provided server. With no imposed milestones or external pacing, I advanced day by day by **developing incrementally**, identifying weaknesses, and adjusting my choices as needed.

Despite this autonomy, I was not isolated. Whenever I encountered a technical or conceptual difficulty, I could **ask questions** to Professor Tajima or other lab members and always received

**clear and helpful answers**. I also regularly exchanged ideas with another international student working on a similar project. Our discussions often turned into *friendly duels* between our respective AIs, which pushed us to refine our models and try new strategies. These interactions were both **technically stimulating** and **personally enriching**.

This experience taught me how to **approach a technical problem from scratch**, to analyze potential solutions, and to adapt based on observed results. It also helped me discover a real interest in **artificial intelligence**, a field I was previously unfamiliar with. I learned everything by testing ideas myself, without relying on pre-existing code or off-the-shelf models. This **hands-on approach** not only strengthened my programming skills but also helped me grow in **autonomy and confidence**.

Ultimately, the goals of this internship went far beyond simply developing a performing AI. It was about **learning how to learn**, exploring a new technical field, and working effectively without constant supervision. The **freedom** I was given allowed me to truly **take ownership** of the project and made the experience **deeply valuable**, both technically and personally.

# Technical Work and Implementation

# 5 Understanding the Game and Defining the Problem

Before diving into the development of the artificial intelligence, it was essential to fully understand the mechanics of the game *Penguin Party*. This card game involves players building a pyramid using their cards, following a specific rule : a card can only be placed on top of two other cards of the **same color**. The objective is simple : get rid of all your cards. If a player cannot make a move, they must pass, and the winner is the one with no cards left, or the fewest cards remaining at the end.

Despite its apparent simplicity, the game hides a surprising level of strategic depth. It is based on **partial information** : each player only knows their own hand. As a result, the AI must learn to make decisions in an environment full of uncertainty. This means developing a strategy that balances risk, adapts dynamically, and makes the most out of limited knowledge.

The game's dynamics are also challenging from a decision-making perspective. The available actions evolve continuously based on the state of the pyramid. This forces the AI to **re-evaluate its options at every turn**. Moreover, there is no direct numerical score to optimize during the game—victory is based on minimizing the number of cards left, a metric that requires longer-term strategic planning rather than short-term gain.

From a technical point of view, I was required to use the **Java-based server** provided by my professor. This server handled the entire game logic and communication between agents. It acted as a black box : I could not access the opponent's moves, their hand, or any internal state beyond what my own AI was doing. Additionally, I was allowed to use only **Java or Python**. I chose Python for its ease of use and for its powerful libraries such as `NumPy` and `pandas`, which proved useful for internal data processing and potential learning strategies.

Before implementing any agent, I spent time analyzing how to represent the game's state computationally. I modeled the pyramid as a 2D matrix, where each cell corresponds to a placed card or an empty slot. The player's hand was represented as a simple list of remaining cards. The agent's primary objective was to **minimize the number of cards left** in its hand by the end of the game, regardless of what the opponent did.

This initial work laid a solid foundation for the development phase. It helped define the constraints of the problem clearly and clarified the type of reasoning my AI would need to perform. The next step was to experiment with different algorithmic approaches to turn this theoretical analysis into effective in-game decisions.

# 6 Choice and Implementation of the First AI

At the beginning of my work on the artificial intelligence for *Penguin Party*, I explored **various approaches** to identify the one best suited to the specific mechanics of the game. I decided to start with a **supervised machine learning** method, which was relatively simple to implement, aiming to create an AI capable of playing **efficiently**.

To achieve this, it was necessary to design a *clear numerical representation* of the game state. Unlike many other games, *Penguin Party* does not involve a deck or draws : the cards are represented solely by **colors**. Therefore, I developed an encoding where each position on the *pyramid* was associated with a specific color, and the player's *hand state* was also taken into account. This **fixed representation** allowed the model to be fed with *structured* and *interpretable* data.

The model itself, a **neural network**, was trained using feedback based on the *final outcome* of each game. Instead of assigning a reward to each move, the reward was given **only at the end** of the game depending on whether it was won or lost. This strategy guided the AI towards sequences of moves that were *globally effective* rather than focusing on short-term individual moves.

This **training approach** enabled me to optimize the AI's behavior to reach a level sufficient to consistently beat a *random AI* and to compete against a human player. I also carefully managed the **amount** and *diversity* of training data to prevent *overfitting* and to ensure **robustness** against various game configurations.

In parallel, I attempted to implement a **Q-Learning** based approach, a reinforcement learning algorithm. This method, which consists of learning the *value of actions* depending on states in order to maximize **cumulative reward**, seemed promising. However, the very large **state space** combined with the challenge of properly balancing *exploration* of new actions and *exploitation* of the best known strategies made learning difficult. Additionally, the difficulty in effectively representing the game state in this context limited the AI's ability to converge towards **performant strategies**.

Tests based on the *average number of cards remaining in the hand* after 100 victories demonstrated that the AI based on supervised machine learning was **more stable** and **effective** than the one developed with Q-Learning, which failed to reach a comparable level. These results highlighted the **technical challenges** related to applying reinforcement learning methods in environments where the *dimensionality of states* is high and where the available training data is limited.

# 7 Evaluation, Testing, and Refinement

To evaluate the effectiveness of the artificial intelligence developed for *Penguin Party*, I implemented a rigorous and consistent testing method. The main metric used was the **average number of remaining cards after 100 victories**. This choice was natural, as the main goal of the game is not only to win, but to **get rid of all one's cards**. By focusing solely on winning games, I was able to measure the overall efficiency of the AI more precisely, avoiding bias caused by accidental defeats or exceptional game situations.

The tests were conducted by making my AI play against a **random AI**, used as a baseline reference, and also against AIs developed by other Japanese students. These matchups allowed me to assess the agent's ability to deal with **diverse playstyles** and to adapt to unpredictable situations.

Throughout the experiments, I attempted to improve the AI's performance by modifying the **reward strategy**. The idea was to introduce **intermediate rewards** based on specific strategic goals during the game, in order to guide the learning process more finely and avoid relying solely on the final outcome. Unfortunately, these adjustments did not yield the expected results. The AI tended to lose track of long-term objectives, focusing instead on short-term gain, or misinterpreting the intermediate signals, which led to degraded overall performance. After several unsuccessful attempts, I decided to **return to a single final reward**, which proved to be more stable and effective.

Matches against other students' AIs played a crucial role in the refinement process. Unlike the random AI, these opponents followed sometimes very specific strategies, which highlighted certain **weaknesses** in my model : lack of decision-making variety, difficulty adapting to aggressive or unexpected playstyles, and poor card placement in edge-case scenarios. These observations were invaluable for making targeted adjustments, especially regarding how the game state was **represented and interpreted** by the model.

Ultimately, this testing and iterative refinement phase transformed a merely functional agent into a truly competitive AI. Far from being linear, the process was marked by trial and error, backtracking, and steady progress at each step. It taught me how to **analyze empirical results**, to **question technical choices**, and to adopt a **methodical and critical approach** when fine-tuning an autonomous system.

# 8 Limits, Learnings, and Future Perspectives

Despite the satisfactory results obtained with the development of the AI for *Penguin Party*, several **technical limitations** restricted the implementation of more advanced strategies. The **Java-based server** provided by my professor handled all the game logic and served as a rigid interface, offering no access to crucial information such as the opponent's hand, their previous moves, or even their card placements. This constraint made it impossible to incorporate advanced techniques like opponent modeling or move prediction. Moreover, the communication protocol was very strict, providing little to no feedback in case of errors, which made debugging and testing significantly more difficult.

Nevertheless, this project turned out to be a highly **formative experience**. The autonomous nature of the task forced me to adopt a deeply **empirical** approach. Without any preexisting solution or framework, I had to make my own technical decisions, test various hypotheses, observe outcomes, and iteratively improve my design. This methodology strengthened my ability to analyze complex behaviors, to adapt when facing obstacles, and to draw clear conclusions from hands-on experimentation. I also realized that in some situations, a **simple but well-designed** system can outperform more sophisticated approaches that lack stability or training efficiency.

Looking ahead, several **improvements** and directions for future work could be considered. One promising idea would be to develop a **hybrid AI**, combining rule-based logic with machine learning. For instance, a fixed rule system could govern early-game decisions, while a trained model takes over for more complex mid- or late-game scenarios. Reinforcement learning could also be revisited, this time with a better **state representation** and a more powerful architecture, such as convolutional or recurrent neural networks, to better capture the spatial and temporal structure of the game. Finally, incorporating **probabilistic strategies**—such as estimating the opponent's hand based on observed moves—could offer a significant edge in real matchups.

In summary, this project exposed me to realistic challenges and gave me the opportunity to explore multiple paths for solving a non-trivial problem. The **lessons learned**, both technical and methodological, will undoubtedly be valuable in future AI projects and beyond.

# Personal and Professional Experience During the Internship

# 9 Cultural and Social Immersion

My internship in Japan was not only a professional experience but also a deep *cultural and social immersion*. From the very beginning, I found myself in a completely different environment where the social codes, behaviors, and everyday habits were unlike anything I had previously known. This required a period of *progressive adaptation*, as well as a genuine sense of *open-mindedness*.

I gradually discovered the richness of Japanese culture through my interactions with both students and professors at the **National Institute of Technology, Gifu College**. I quickly realized how important values such as **respect**, **punctuality**, and **discretion** are in Japanese society. Learning to adjust my behavior accordingly helped me develop a stronger *intercultural sensitivity* and a deeper understanding of local customs.

Beyond the academic setting, I had the opportunity to participate in several activities organized by the school, such as a **large barbecue**, the weekly **Talk Café** (designed to foster intercultural and language exchanges), and a **sports day**. One of the most memorable moments for me was a **friendly football match** shared with Japanese students. Despite cultural and linguistic differences, the universal language of sports helped us connect in a natural and sincere way.

These social interactions were a key factor in my integration. Communication was not always easy, but we managed to understand each other through a mix of **English**, a few words in **Japanese**, and even gestures or simple smiles. This motivated me to learn some basic Japanese vocabulary, further strengthening my desire to connect with others.

My immersion, however, extended far beyond the school walls. During weekends, I had the chance to **travel across Japan**, exploring both major cities and smaller towns. I visited the Kansai region — **Kyoto, Osaka, Nara**, and **Kobe** — and also discovered more local destinations like **Seki** and **Gero**, which offered a more authentic view of rural Japan. One particularly unique experience was traveling to **Tokyo** by overnight bus, which gave me a firsthand glimpse into the fast-paced rhythm and vibrant atmosphere of the capital.

Living in Japan also brought moments of *introspection and solitude*. Being far from familiar surroundings pushed me to **recenter myself** and face some of my personal fears, including the fear of not being understood or accepted. Yet through this experience, I gained a better understanding of who I am, increased my self-confidence, and broadened my worldview in meaningful ways.

# 10 Personal Development and Transversal Skills

During my internship, I did not encounter any particular difficulty adapting to the environment. From the very beginning, I felt at ease in the work setting. The only constraint was the fixed working hours, from **9 :30 a.m. to 4 :10 p.m.**, but within that framework, I was granted full **autonomy**. This freedom pushed me to structure my own time and tasks, without any imposed deliverables or deadlines.

This autonomy played a central role in my development. When I faced technical challenges such as a bug where my AI did not make a move when it was the first player I was able to identify and solve the issue independently. I relied on documentation, code inspection, and experimentation to find a solution. This greatly improved my ability to **learn by doing** and to find answers without direct assistance.

In addition to technical problem-solving, I also grew on a more personal level. Traveling alone in large cities like **Tokyo** and **Osaka**, I had to plan everything by myself, manage transportation, and adapt to unfamiliar situations. These solo experiences, although outside the professional scope, taught me how to be more **independent**, make decisions on my own, and stay focused in unfamiliar environments.

The combination of academic responsibilities and daily life challenges led me to develop essential **transversal skills** : *self-management*, *initiative*, *adaptability*, and *resilience*. These skills, acquired progressively, are not limited to technical tasks—they will be valuable throughout my academic path and future professional experiences.

# 11 Professional Impact and Perspectives

This internship had a significant impact on the way I perceive the professional world, even though it did not provide me with definitive answers about my future career path. Above all, it helped me realize that I am capable of carrying out a complex technical project independently, within an international environment. This experience has been deeply formative, giving me greater self-confidence in my ability to manage a project from start to finish without constant supervision.

From a technical perspective, I developed concrete skills in the field of Artificial Intelligence, particularly in **machine learning** and **Q-learning**. More importantly, I learned how to properly conduct *independent research*, seek out information, and resolve issues through self-directed problem solving. This type of hands-on learning, grounded in real-world scenarios, has allowed me to grow in a much more practical and meaningful way than any theoretical course could

offer.

Although this experience has proven that I can evolve in the field of computer science, it did not entirely convince me to pursue a long-term career in it. I still have doubts, but they are now shaped by real-world exposure, which already represents a major step forward in my reflection compared to where I stood before the internship.

Beyond the academic and technical aspects, this experience also opened up **unexpected professional opportunities**. I received multiple job offers from Japanese individuals I met during the internship — including from a **high-end restaurant** and a **private tutoring company**. These proposals, although surprising, reflect the unique doors that cultural immersion can open and have made me seriously consider returning to Japan in the future, either to work or to live there for an extended period.

Finally, I plan to maintain contact with several people I met during my stay, both in a professional and personal context. These relationships represent a valuable network that could, in the future, help me better understand the workings of professional life in Japan — or even support me in new international endeavors.

# Conclusion

My internship at the **National Institute of Technology, Gifu College** was a true *personal and professional transformation*. More than just a technical project, it was a complete immersion into a new cultural and professional environment, presenting me with multiple, sometimes unexpected challenges.

On the technical side, I had to learn to operate in a context with limited resources, sparse documentation, and where **adaptability** was key. Designing a competitive artificial intelligence for *Penguin Party* required not only programming and machine learning skills, but also great **rigor** in problem analysis and the ability to **test, debug, and iterate** autonomously. This experience demonstrated the importance of **patience**, scientific curiosity, and especially **independence** — qualities essential for any developer or researcher.

Beyond the technical aspects, this internship was a deeply *human adventure*. Being immersed in Japanese society, even briefly, offered me a new perspective on intercultural relationships, managing solitude, and quickly adapting without losing one's identity. The moments shared with local students, as well as traveling across diverse regions of Japan, were opportunities for **openness** and personal enrichment. I learned to better manage *uncertainty* and to cultivate **resilience** in the face of both professional and personal difficulties.

This internship also became a time for deep reflection about my future. It showed me that I am capable of successfully completing a complex project in a demanding field, yet it also revealed that my professional aspirations still need clarification. The job offers I received from Japanese companies open up an international horizon I had not fully considered, and this experience encourages me to envision a career that could extend beyond borders. In that sense, this internship is a **starting point**, not a conclusion.

Ultimately, this experience provided a rare combination of technical, human, and cultural lessons. It gave me confidence in my abilities, while emphasizing the importance of staying **curious**, adaptable , and **open-minded**. What I take away from it most is a renewed desire to keep learning, exploring, and pushing myself, whatever paths I choose to follow tomorrow.